LESSON 3-3 · IDENTITIES WITH POLYNOMIALS

Choose Your Route: Practice

Algebra II · Choose Your Route

Student ..... **Class** ..... **Date** ..... / ..... / 2026 **Teacher** Mr Malek Thiab

Three routes, one destination. Start anywhere on pages 1 and 2, move between routes whenever you like, and finish with page 3. A route names the task, not you.

PRACTICE · SECURE THE METHOD

Cards 1 to 3: expand using an identity. Cards 4 and 5: factor using an identity. Card 6: expand with a row of Pascal's triangle. Then prove one identity in the box on the right.

1 Expand $(3x+4)^2$

2 Expand $(5x-2)^2$

3 Expand $(2x+7)(2x-7)$

4 Factor $16x^2-81$

5 Factor $x^2+10x+25$

6 Expand with Pascal's row 4 $(x+3)^4$

Working space. Square the first term, double the product of the two terms, square the last term. Row 4 is $1,\ 4,\ 6,\ 4,\ 1$.

Prove it: show that $(a+b)^2-(a-b)^2=4ab$ by expanding both squares. Then say in one line what it tells you about 53^2-47^2 .

FIND THE ERROR

A student expanded a square and then checked the result at $x=2$. Say what went wrong and write the correct expansion.

$(2x-3)^2=4x^2-9$

Check at $x=2$: $(4-3)^2=1,\ \ 4(2)^2-9=7$

Done when... the middle term $2ab$ is written, its sign follows the bracket, and each answer is checked with a number.

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Choose Your Route: Apply and Investigate

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APPLY · USE IT IN CONTEXT

1. A model solar-panel field is a square of side $(x+4)$ m. Write its area as an expanded polynomial. Find the area when $x=16$ without a calculator.

2. A model event hall has 103 rows of 97 seats. Use an identity to find the number of seats without a calculator.

3. Show that $(2n+1)^2-1$ is a multiple of 4 for every whole number n.

4. Expand $(x-2)^5$ using row 5 of Pascal's triangle.

Done when: the identity you used is named, every step stays balanced, and each answer says what it means.

INVESTIGATE · FIND OUT WHY

1. Prove that $(a+b)^2-(a-b)^2=4ab$. Use it to find 53^2-47^2.

2. Show that $(x^2-y^2)^2+(2xy)^2=(x^2+y^2)^2$. Use $x=4$ and $y=1$ to build a Pythagorean triple.

3. Add the numbers in each of rows 0 to 5 of Pascal's triangle. Describe the pattern and explain it by putting $a=b=1$ in $(a+b)^n$.

Done when: your argument starts from a known identity or a full expansion, and each pattern is explained, not just noticed.

SPOTLIGHT · AN IDEA THAT IS MY OWN

A route nobody else used, a question nobody else asked, or a surprise you found:

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Which Route? Judge the Claims

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COMPARE THE ROUTES

Expand $(x+2)^4$ in three ways. Then compare: which was quickest, and which would you trust for $(x+2)^8$?

ROUTE	my working	steps	where a slip could happen
Multiply out repeatedly			
Pascal's triangle			
Binomial theorem with coefficients			

My choice and the evidence for it:

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JUDGE THE CLAIMS

Decide true, false or sometimes true. One counterexample is enough to show a claim is false; a few examples are not enough to show it is true.

A	$(a+b)^2 = a^2 + b^2$ **for all numbers** a **and** b **.**	☐ true ☐ false ☐ sometimes	*My evidence:* *Assumption I checked:* Did I test numbers where neither a nor b is 0 ?
B	**The numbers in row** n **of Pascal's triangle add up to** 2^n **.**	☐ true ☐ false ☐ sometimes	*My evidence:* *Assumption I checked:* Did I check more than one row?
C	$(a-b)^2$ **and** $(b-a)^2$ **are always equal.**	☐ true ☐ false ☐ sometimes	*My evidence:* *Assumption I checked:* Did I expand both, or test only one pair of numbers?

WHAT IF...? PREDICT FIRST, THEN TEST

1. What if the second term is negative, as in $(x-2)^4$? Predict which signs alternate. Then test with $x=1$.

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2. What if the power is 0 , as in $(a+b)^0$? Predict the value and which row of Pascal's triangle it matches.

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3. Write a What if...? question of your own about an identity, predict the answer, and test it.

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Done when... you have compared three routes with evidence, judged each claim with an example or counterexample, and tested at least one What if...? idea.